

COMPLETE ENGINEERING HANDBOOK

MASTER EDITION

ShadowTalk AI Master System Documentation

The definitive encyclopedic guide: System architecture, Turbo Engine heuristics, 30+ native tools, Personal IDE, cryptographic privacy, and complete codebase dictionary.

FOUNDER & LEAD ARCHITECT

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ARCHITECTURE VERSION

v3.0.0 (Unified Agentic Pivot)

COMPILATION STATUS

TypeScript Strict (0 Errors)

PRODUCTION BUILD STATUS

Vite PWA (100% Passed)

Confidential • Master Unified Edition

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VOLUME 1

Architecture & System Foundations

Component hierarchy, lifecycle management, dual-backend hybrid model, and hardware performance profiling.

1 Philosophy, Vision & Evolution

Modern knowledge workers and developers suffer from severe cognitive fragmentation across disparate AI interfaces. Writing prompts in one tab, debugging code in an isolated playground, researching in a search engine, and managing documents in an external cloud produces friction and context loss. **ShadowTalk AI** solves this by synthesizing conversational intelligence, autonomous agents, in-browser code execution, and persistent memory into a single workspace.

DreamState

BACKGROUND AGENT

Executes long-horizon multi-turn tasks while the user is disconnected.

Omniscience

CONTEXT MEMORY

Connects facts across sessions and attached documents via adaptive vector graphs.

Shadow Twin

DIGITAL PERSONA

Reflects user operational style, tone, and logic with public shareable links.

The v3.0.0 Architecture Consolidation

In version 3.0.0, the repository underwent a major architectural pivot. Over 40 disparate legacy standalone pages were consolidated directly into the unified `/chatbot` canvas and modal command decks. This drastically reduced the routing footprint, accelerated code-splitting, and focused the application on agentic autonomy.

2 Application Lifecycle & Component Hierarchy

The application entry point is `src/main.tsx`, which boots critical polyfills, error interceptors, and device performance profiling before mounting the React tree:

Root Boot Sequence

- UUID Polyfill:** Ensures `crypto.randomUUID` is defined even in unauthenticated LAN/IP testing contexts.
- Global Error Interception:** Registers `setupGlobalErrorHandling()` to catch unhandled promise rejections and operational API errors.
- Hardware Tier Detection:** Invokes `applyPerfProfile()` synchronously before CSS paints to calibrate performance tier attributes.
- Chunk Recovery:** Installs `installViteChunkRecovery()` to automatically refresh stale dynamic import hashes after production deployments.
- Idle Warmup:** Defers non-critical WebGPU probing and service worker registration using `window.requestIdleCallback`.

Global Provider Hierarchy

The provider tree wrapped in `src/App.tsx` guarantees end-to-end security, state coordination, and theming:

```
<ErrorBoundary>
  <QueryClientProvider client={queryClient}>
    <ThemeProvider attribute="class" defaultTheme="dark" forcedTheme="dark">
      <TooltipProvider>
        <AuthProvider>
          <StealthKillSwitchProvider>
            <SecurityProvider>
              <ShadowMemoryProvider>
                <AutoImproveProvider>
```

```
<ThemeProvider>
  <CommandPaletteContext.Provider>
    <BrowserRouter>
      <SiteMotionProvider>
        <SitePageShell>
          <AnimatedRoutes />
        </SitePageShell>
      </SiteMotionProvider>
    </BrowserRouter>
  </CommandPaletteContext.Provider>
</ThemeProvider>
</AutoImproveProvider>
</ShadowMemoryProvider>
</SecurityProvider>
</StealthKillSwitchProvider>
</AuthProvider>
</TooltipProvider>
</ThemeProvider>
</QueryClientProvider>
</ErrorBoundary>
```

3 Dual-Backend Hybrid Storage Architecture

ShadowTalk AI employs a dual-tier storage strategy designed to function seamlessly in both 100% offline air-gapped environments and fully connected cloud environments:

Storage Tier	Underlying Technology	Scope & Responsibility
Local-First Layer	IndexedDB (<code>idb</code>), LocalStorage, Web Crypto API	Encrypted personal vault, adaptive turn memories, BYOK API keys, and guest conversation caches. Operates with zero network dependency.
Cloud Persistence	Firebase Firestore, Firebase Auth, Cloud Functions v2, Storage	Multi-device sync, verified user roles, audit trails, and shared LLM proxy quotas.

The Supabase PostgREST Adapter Layer

To avoid rewriting database queries across hundreds of components during backend migrations, ShadowTalk implements a transparent adapter in `src/integrations/firebase/adapter.ts` and `firestore.ts`:

Adapter Mechanics

When components invoke `backend.from('conversations').select('*').eq('user_id', uid).order('created_at', { ascending: false }).limit(20)`:

- The query builder translates SQL-like clauses into Firestore `query()`, `where()`, `orderBy()`, and `limit()` constraints.
- For operators unsupported natively in Firestore (e.g. `ilike`, arbitrary nested `is` nulls, complex OR groups), the adapter fetches matching document candidates and applies an in-memory post-filter via `matchesLocal()`.
- Timestamps formatted as Firestore `Timestamp` objects are automatically normalized into ISO-8601 strings, maintaining signature parity.

4 Hardware Performance & Low-End Degradation

Ensuring smooth 60 FPS operation across mobile devices, low-cost Chromebooks, and high-end workstations is handled by `src/lib/perf/devicePerfTier.ts`:

Tier Detection Factors

- VRAM & Max Buffer Size:** Probed via WebGPU adapter limits.
- Logical Cores:** `navigator.hardwareConcurrency`.
- Device Memory:** `navigator.deviceMemory` (GB).
- Power Preference:** Battery status API integration.

Automated Degraded Modes

- Tier 0 (Low-End):** Disables Three.js 3D canvas, turns off backdrop-filter blurs, simplifies spring transitions to standard CSS fades.
- Tier 1 (Standard):** Enables subtle motion, standard shadows.
- Tier 2 (High-End):** Full WebGL/Three.js pulse orbs, ambient canvas glow, real-time particle rendering.

VOLUME 2

AI Engines & Agentic Ecosystem

Turbo Engine complexity routing, sovereign on-device
WebGPU runtimes, 30+ native tools, cognitive loops, and
adaptive memory.

1 The Turbo Engine: Dynamic Model Routing

Dispatching every user inquiry to heavy frontier models introduces unnecessary token expenditure and unacceptable latency for simple greetings. Conversely, routing complex system design questions to lightweight models leads to reasoning failures. ShadowTalk solves this through the **Turbo Engine** in `src/lib/turbo/`.

Heuristic Complexity Scoring (`analyzeComplexity`)

Every message array is evaluated before dispatch based on semantic keywords and character payload volume:

- **Low Complexity:** Matches common greetings (`hi`, `hello`, `thanks`), formatting instructions (`tldr`, `fix typo`), or prompt lengths under 100 characters. Automatically routed to **Groq** (`llama-3.1-8b-instant`), achieving token generation speeds exceeding 300 tokens/sec.
- **Medium Complexity:** General conversational turns, creative synthesis, and standard queries routed to **Groq** (`llama-3.3-70b-versatile`).
- **High Complexity:** Triggered by architectural terms (`refactor`, `architect`, `security`, `typescript`, `sql`, `synthesize`) or contexts exceeding 4,000 characters. Escalated to **OpenAI** (`gpt-4o`) or **Anthropic** (`claude-3-5-sonnet`).

Sovereign & On-Device Inference

ShadowTalk provides full support for air-gapped and zero-leak operations:

- **WebGPU Local Runtime:** `src/lib/webgpuRuntime.ts` probes device VRAM. On compatible GPUs, models like Gemma-2B run client-side using WebAssembly and WebGPU compute shaders via `@mlc-ai/web-llm`.
- **Ollama Desktop Bridge:** When running locally, ShadowTalk communicates with local daemon endpoints on `http://localhost:11434` without passing data to external servers.
- **BYOK (Bring Your Own Key):** Stored strictly in local encrypted storage, authorizing direct edge client-to-provider streaming.

2 The 30+ Native Tool Registry

The tool orchestration pipeline (`src/hooks/useToolOrchestrator.ts` and `executeShadowTool.ts`) provides conversational execution of native actions:

Tool Identifier	Target Handler	Capability Description
<code>web_search</code>	Cloud Function <code>web-search</code>	Queries live web indexes, retrieves snippets, and synthesizes multi-source citations.
<code>deep_research</code>	Turbo Engine Prompt Graph	Executes multi-step recursive web research, formulating follow-up queries and final reports.
<code>image_generator</code>	Pollinations / Flux API	Generates photorealistic images with custom seeds, aspect ratios, and styles.
<code>image_decoder</code>	Multimodal Vision Bridge	Performs OCR, document inspection, and diagram reasoning on base64 image attachments.
<code>shadow_browser</code>	Firecrawl Scraping Gateway	Fetches live JS-rendered URLs, stripping ads and converting DOM to clean Markdown.
<code>code_executor</code>	WebContainer Node.js Engine	Spawns processes, executes scripts in sandboxed WebAssembly, and streams terminal output.
<code>presentation_builder</code>	Client <code>pptxgenjs</code> Pipeline	Creates PowerPoint slides, bullet points, styling, and exports directly to <code>.pptx</code> .
<code>database_query</code>	Backend Function <code>postgres-query</code>	Runs schema inspection and sanitized SQL read queries, formatting rows into JSON tables.
<code>shadowspectre</code>	Cyber Defense Agent	Performs vulnerability assessments, security header audits, and exploit path validation.

3 Cognitive Loops & Swarm Consensus

Beyond single-turn tool calls, ShadowTalk implements multi-step autonomous cognitive loops:

The Cognitive Loop Framework (CognitiveLoopPanel.tsx)

When activated, the agent enters an autonomous 4-stage execution cycle:

1. **Plan:** Breaks high-level user directives into structured, ordered sub-tasks.
2. **Execute:** Sequentially dispatches sub-tasks to tools or reasoning models.
3. **Reflect:** Evaluates tool execution output against the original user criteria. If an error or partial result occurs, the agent self-corrects.
4. **Synthesize:** Blends all sub-task artifacts into an exhaustive final output.

Multi-Model Consensus (useMultiModelConsensus.ts)

For high-stakes decision making, ShadowTalk broadcasts the identical prompt simultaneously across three distinct model families (e.g. Groq Llama-3.3, OpenAI GPT-4o, and Google Gemini 2.0). The responses are analyzed for semantic convergence, surfacing agreements, edge-case discrepancies, and a calculated confidence rating.

4 Adaptive Memory Engine

Persistent user context is managed by `src/lib/memory/adaptiveMemory.ts`. Unlike dumb chat transcripts, Adaptive Memory scores and categorizes high-signal statements:

Turn Scoring Heuristics

- **Identity Signals:** Statements matching `my name` is, `i am`, `i build` gain +0.40 confidence.
- **Preference Signals:** Matches on `i prefer`, `always`, `never` gain +0.35 confidence.
- **Decision Signals:** Matches on `we decided`, `going with` gain +0.30 confidence.
- **Task Signals:** Matches on `build`, `ship`, `deploy` gain +0.20 confidence.

IndexedDB Storage Schema

- **Store Name:** `facts` in database `shadowtalk_memory_db`.
- **Indexes:** Indexed on `by-confidence` and `by-lastUsedAt`.
- **Auto-Pruning:** When facts exceed capacity, least-recently-used (LRU) low-confidence facts are purged to avoid context bloat.

VOLUME 3

Developer Studio & Security Vault

In-browser virtual Node.js OS via StackBlitz WebContainer,
Monaco Editor, PBKDF2 + AES-GCM encryption, and Stealth
Network Guard.

1 Personal IDE & WebContainer Runtime

ShadowTalk features a complete in-browser software development suite (`src/components/chat/PersonalIDE.tsx` , 103 KB) and `src/lib/webcontainer/engine.ts` . Developers can write, compile, run, and preview full-stack Node.js projects without installing local runtimes:

WebContainer Virtual Operating System

Using WebAssembly and browser `SharedArrayBuffer` primitives, StackBlitz WebContainer boots a headless Node.js virtual kernel directly in browser memory:

- **File System Tree:** The workspace converts project `FileNode[]` structures into a hierarchical `FileSystemTree` mounted via `instance.mount(tree)` .
- **Command Execution:** Spawns processes (e.g. `npm install` , `node index.js`) via `instance.spawn()` , streaming stdout and stderr to a terminal interface.
- **Live Dev Server Preview:** Detects open server ports (e.g. port 3000 or 5173) and displays an embedded, hot-reloading iframe preview inside the workspace canvas.

Monaco Editor Integration

The code editor leverages Microsoft's Monaco Editor engine with support for multi-tab navigation, syntax highlighting, diff viewing, line bookmarking, and automatic formatting across TypeScript, JavaScript, Python, HTML, CSS, SQL, and JSON.

2 End-to-End Cryptography (E2EE)

User privacy in ShadowTalk is enforced at the mathematical level via `src/lib/e2e-encryption.ts` using standard Web Crypto API primitives:

Key Derivation (PBKDF2)

The user's master vault password never leaves the device. Keys are derived using PBKDF2 with **100,000 iterations** of SHA-256 and a cryptographically secure 16-byte random salt generated via `crypto.getRandomValues()` .

Authenticated Encryption (AES-GCM)

Plaintext payloads are encrypted with **AES-256-GCM** using a unique 12-byte initialization vector (IV) per message. The ciphertext and authentication tag ensure tamper detection and confidentiality on cloud servers.

3 Stealth Kill Switch & Network Guard

For sensitive operational security (OPSEC), ShadowTalk includes an active network kill switch in `src/lib/stealthNetworkGuard.ts` :

Active Network Monkey-Patching

When Stealth Mode is engaged via `StealthKillSwitchContext.tsx` :

- `window.fetch` and `XMLHttpRequest.prototype.open` are dynamically overridden.
- Outgoing URLs are checked against a strict whitelist allowing only `blob:` , `data:` , and verified local origins (`localhost` , `127.0.0.1`).
- Any outbound telemetry, cloud sync, or external API call is immediately terminated, triggering a blocked request counter in the UI.

4 Biometric Authentication & Threat Intelligence

Access control to encrypted local vaults is secured via modern web authentication standards:

- **WebAuthn Biometrics (useBiometricAuth.ts):** Integrates platform authenticators (Face ID, Touch ID, Windows Hello) to unlock local AES keys without retyping master passwords.
- **ShadowSpectre Cyber Suite:** Implemented in `src/lib/cyber/shadowspectre/`. Includes specialized security reasoning personas capable of parsing vulnerability reports, auditing HTTP response headers, triaging CVE alerts, and validating defensive network policies.

5 Cloud Security Rules & Access Controls

Cloud persistence in Firebase is fortified through declarative rules enforced at the database kernel level:

Firestore Security Rules (firestore.rules)

```
service cloud.firestore {
  match /databases/{database}/documents {
    function signedIn() { return request.auth != null; }
    function isAdmin() {
      return signedIn() &&
        exists(/databases/{database}/documents/user_roles/{request.auth.uid}) &&
        get(/databases/{database}/documents/user_roles/{request.auth.uid}).data.role == 'admin';
    }
    function ownsExisting() { return signedIn() && resource.data.user_id == request.auth.uid; }
    function ownsIncoming() { return signedIn() && request.resource.data.user_id == request.auth.uid; }

    // User Roles: Admin managed only
    match /user_roles/{uid} {
      allow read: if signedIn() && (request.auth.uid == uid || isAdmin());
      allow write: if isAdmin();
    }

    // Default per-user isolation
    match /{collection}/{doc} {
      allow read: if ownsExisting() || isAdmin();
      allow create: if ownsIncoming();
      allow update, delete: if (ownsExisting() && ownsIncoming()) || isAdmin();
    }
  }
}
```

Cloud Storage Security Rules (storage.rules)

File uploads are restricted to `/users/{uid}/*` folders. Uploads enforce ownership verification (`request.auth.uid == uid`), maximum payload caps of 50 MB, and explicit MIME-type filtering to prevent executable payload uploads.

VOLUME 4

Enterprise SaaS & File Dictionary

Monetization architecture, Stripe integration, white-label
tenanting, exhaustive directory catalog, and deployment
runbook.

1 SaaS Monetization & Token Economics

ShadowTalk incorporates an enterprise-ready monetization infrastructure spanning monthly recurring subscriptions, micro-transaction token billing, and add-on services (`src/lib/pricingCatalog.ts` and `src/lib/stripe.ts`):

Plan Tier	Pricing	Daily Allowances	Feature Entitlements
Free Tier	\$0 / month	50 messages / day, 3 deep research, 5 images	Standard Groq access, local WebGPU inference, community support.
Pro Tier	\$10 / month	Unlimited messages, 20 deep research, 20 images	Turbo Engine auto-routing, full code executor, priority queue.
Premium Tier	\$15 / month	Unlimited messages, 50 deep research, 50 images	All agents, Cognitive Loop, multi-model consensus, voice copilot.
Elite Tier	\$20 / month	Unlimited everything (fair-use capped)	White-label branding, stealth vault, dedicated tenant SLA, phone support.

Shadow Credits Micro-Billing (useShadowCredits.ts)

To support pay-per-use workflows without forced recurring subscriptions, users can purchase credit packs. High-compute operations dynamically deduct credits: Image Generation (2 credits), Deep Research Synthesis (5 credits), Voice Copilot Session (1 credit/minute).

2 Enterprise Tenancy & Viral Growth

The workspace includes enterprise customization and community virality hooks:

- **White-Label Branding (WhiteLabelBranding.tsx):** Allows enterprise customers to override application logos, navigation titles, primary theme colors, and custom subdomains.
- **Viral Referral Engine (ReferralProgram.tsx):** Captures query parameters via `useReferralTracking.ts`, tracks conversion funnels, and automatically grants free tier upgrades upon successful friend activations.
- **Ethical Behavioral Intelligence (useProactiveAI.ts):** Evaluates visitor hesitation, dwell time, and interaction frequency to offer contextual guidance while adhering to strict privacy guardrails (`PROACTIVE_ETHICS`).

3 Exhaustive Codebase Directory Dictionary

A structured index of the primary repository directories and architectural responsibilities:

Directory Path	Key Sub-Modules	Functional Domain
src/components/chat/	PersonalIDE.tsx , ShadowBrowser.tsx , CognitiveLoopPanel.tsx	Primary interactive chat canvas, modal panels, code canvases, and tool inspectors.
src/lib/turbo/	turboEngine.ts , modelRouter.ts , turboProviders.ts	Intelligent complexity routing algorithm and multi-provider streaming client.
src/lib/memory/	adaptiveMemory.ts , promptInjector.ts	IndexedDB vectorized turn-scoring and contextual prompt memory injection.
src/lib/cyber/	shadowspectre/ (client, prompts, router, scope)	Autonomous security evaluation, vulnerability scanning, and threat intelligence.
src/integrations/firebase/	adapter.ts , firestore.ts , auth.ts , storage.ts	PostgREST query adapter, authentication wrappers, and Firestore synchronization.
functions/src/	index.ts (shared pool, audio mock, drive)	Firebase Cloud Functions v2 handling server-side secrets and rate-limited fallbacks.

4 Build, Testing & Deployment Runbook

Standard operating procedures for compiling, testing, and distributing ShadowTalk AI:

Verification Commands

```
# 1. Static Type Checking (Clean: 0 errors)
npx tsc --noEmit

# 2. Automated Test Suite (29/31 suites passing)
npm test

# 3. Production Vite Compilation
npm run build
```

Deployment Environments

- **Firebase Web Hosting:** Deployed via `firebase deploy --only hosting`. Configured with strict security headers (nosniff, strict-origin-when-cross-origin).
- **PWA Service Worker:** Generated via `vite-plugin-pwa` with Workbox precaching 218 static assets for instant offline launches.
- **Mobile Packaging (Capacitor):** `capacitor.config.ts` defines native application IDs (`com.shadowtalk.ai`), splash screen durations, dark status bar tokens, and keyboard resize behaviors for iOS and Android builds.

Audited Project Health Summary

The project demonstrates exceptional engineering discipline: 100% clean TypeScript compilation with zero type errors, robust code splitting across vendor libraries, complete cryptographic isolation for sensitive data, and seamless fallback to on-device sovereign WebGPU execution.